

Topic B9

Transport in animals

Concept guide · Worked application · Cambridge-style practice · Fully explained answers

What this guide teaches

A complete explanation of the syllabus concepts, biological mechanisms, common misconceptions, data interpretation and original exam-style practice for this topic.

Syllabus map

B9.1	Circulatory systems
B9.2	Heart
B9.3	Blood vessels
B9.4	Blood and lymph

How to study: retrieve definitions, cover the application, then complete the practice paper without using the notes.

1. Double circulation

Humans have a closed double circulation: heart to lungs and back, then heart to body and back. Separation maintains low pulmonary pressure to protect lungs and high systemic pressure for rapid delivery. Fish have a single circulation.

Examiner insight

Trace a complete route using chambers, vessels and gas content.

2. Heart function

The right side pumps deoxygenated blood to lungs; the left side pumps oxygenated blood to the body. Valves prevent backflow. The left ventricle has the thickest wall because it produces high systemic pressure. Coronary arteries supply cardiac muscle.

Examiner insight

Do not say the heart adds oxygen; gas exchange occurs in lungs.

3. Blood vessels

Arteries carry blood away at high pressure and have thick elastic muscular walls. Veins return blood at lower pressure, have wider lumens and valves. Capillaries have one-cell-thick walls, narrow lumens and extensive networks for short diffusion distance and large area.

Examiner insight

Define artery and vein by direction, not oxygen content.

4. Blood components

Red cells transport oxygen using haemoglobin and are biconcave, flexible and anucleate. White cells defend by phagocytosis or antibody production. Platelets initiate clotting. Plasma transports cells, nutrients, hormones, carbon dioxide, urea and heat. Tissue fluid enables exchange with cells.

Examiner insight

Link each cell adaptation to its transport or defence role.

Worked application

Situation: A coronary artery becomes blocked.

Prediction: Cardiac muscle may die and heart output fall.

Reason: Reduced oxygen delivery limits aerobic respiration in cardiac muscle.

Exam technique

High-value habits

Define terms precisely, describe visible patterns before explaining them, link structure to function, use data values as evidence, identify variables explicitly and answer the command word directly.

Retrieval checklist

- Recall five key definitions without notes.
- Explain one process as a linked cause-and-effect chain.
- Interpret the data set and identify a limitation.
- Complete the timed practice and mark only after finishing.

Practice Paper

Topic B9

ransport in animals

Instructions

Answer all questions. Select one option for multiple choice. Use values from tables where relevant. Write explanations on the lines. Individual parts carry 1-3 marks.

Candidate name:

Class / date:

Total marks: 50

Suggested time: 60 minutes.

Section A · Multiple choice

Choose the best answer for each question.

1. Which statement best defines artery? [1]
 - A a blood vessel carrying blood away from the heart
 - B a vessel carrying blood toward the heart
 - C a microscopic vessel with a one-cell-thick wall for exchange
 - D a red-cell protein that combines reversibly with oxygen
2. Which statement best defines vein? [1]
 - A a microscopic vessel with a one-cell-thick wall for exchange
 - B a vessel carrying blood toward the heart
 - C a red-cell protein that combines reversibly with oxygen
 - D a vessel supplying the heart muscle with oxygenated blood
3. Which statement best defines capillary? [1]
 - A a red-cell protein that combines reversibly with oxygen
 - B a vessel supplying the heart muscle with oxygenated blood
 - C a microscopic vessel with a one-cell-thick wall for exchange
 - D a blood vessel carrying blood away from the heart
4. Which statement best defines haemoglobin? [1]
 - A a vessel supplying the heart muscle with oxygenated blood
 - B a blood vessel carrying blood away from the heart
 - C a vessel carrying blood toward the heart
 - D a red-cell protein that combines reversibly with oxygen
5. Which statement best defines coronary artery? [1]
 - A a vessel supplying the heart muscle with oxygenated blood
 - B a blood vessel carrying blood away from the heart
 - C a vessel carrying blood toward the heart
 - D a microscopic vessel with a one-cell-thick wall for exchange
6. Which statement about double circulation is correct? [1]
 - A A vessel that always carries oxygenated blood.
 - B Humans have a closed double circulation: heart to lungs and back, then heart to body and back.
 - C A vessel defined by deoxygenated blood.
 - D A high-pressure vessel with valves.

7. Which statement about heart function is correct? [1]
- A A vessel defined by deoxygenated blood.
 - B A high-pressure vessel with valves.
 - C The right side pumps deoxygenated blood to lungs; the left side pumps oxygenated blood to the body.
 - D An antibody in plasma.
8. Which statement about blood vessels is correct? [1]
- A A high-pressure vessel with valves.
 - B An antibody in plasma.
 - C The vessel carrying blood from heart to lungs.
 - D Arteries carry blood away at high pressure and have thick elastic muscular walls.
9. Which statement about blood components is correct? [1]
- A Red cells transport oxygen using haemoglobin and are biconcave, flexible and anucleate.
 - B An antibody in plasma.
 - C The vessel carrying blood from heart to lungs.
 - D A vessel that always carries oxygenated blood.
10. Which prediction is most appropriate when a coronary artery becomes blocked? [1]
- A A vessel that always carries oxygenated blood.
 - B A high-pressure vessel with valves.
 - C Cardiac muscle may die and heart output fall.
 - D The vessel carrying blood from heart to lungs.

Section B · Structured questions

11 Biologists use precise vocabulary when studying transport in animals.

(a) Define artery. [2]

(b) Define vein. [2]

(c) Distinguish capillary from the incorrect idea: “a high-pressure vessel with valves”. [2]

(d) State two important features of double circulation. [2]

[Total: 8]

12 A coronary artery becomes blocked.

(a) State the most likely biological outcome. [2]

(b) Explain the outcome using a linked biological mechanism. [3]

(c) Predict what would happen if the relevant change were reversed. [2]

(d) State one assumption made in this prediction. [1]

[Total: 8]

13 A student records beats per minute under three conditions.

condition	beats per minute
rest	68
moderate exercise	112
intense exercise	168

(a) Calculate the difference between the highest and lowest recorded values. [2]

(b) Describe the pattern in the results. [2]

(c) Explain the pattern using biological knowledge. [3]

(d) State one limitation of drawing a conclusion from these data. [1]

[Total: 8]

14 A student plans to measure recovery of pulse after exercise.

(a) Identify the independent variable. [1]

(b) Identify the dependent variable. [1]

(c) State two control variables. [2]

(d) Describe how to obtain reliable results across a suitable range. [3]

(e) Suggest one specific improvement. [1]

[Total: 8]

15 Apply your knowledge of transport in animals to an unfamiliar context.

(a) Define haemoglobin. [2]

(b) Explain the biological importance of blood vessels. [3]

(c) Correct this statement: “the vessel carrying blood from heart to lungs”. [2]

(d) State one further question that could be investigated. [1]

[Total: 8]

Worked solutions

Remove or fold back this section while completing the practice paper.

Marking routine

Mark each part separately. Award points for accurate biology, use of evidence and linked reasoning, then diagnose the first error.

Solutions · Multiple choice

- 1 A. A blood vessel carrying blood away from the heart.
- 2 B. A vessel carrying blood toward the heart.
- 3 C. A microscopic vessel with a one-cell-thick wall for exchange.
- 4 D. A red-cell protein that combines reversibly with oxygen.
- 5 A. A vessel supplying the heart muscle with oxygenated blood.
- 6 B. Trace a complete route using chambers, vessels and gas content.
- 7 C. Do not say the heart adds oxygen; gas exchange occurs in lungs.
- 8 D. Define artery and vein by direction, not oxygen content.
- 9 A. Link each cell adaptation to its transport or defence role.
- 10 C. Reduced oxygen delivery limits aerobic respiration in cardiac muscle.

Solutions · Structured questions

11. (a) a blood vessel carrying blood away from the heart [2]. (b) a vessel carrying blood toward the heart [2]. (c) a microscopic vessel with a one-cell-thick wall for exchange; not a high-pressure vessel with valves [2]. (d) Credit any two accurate features from the guide: Humans have a closed double circulation: heart to lungs and back, then heart to body and back. Separation maintains low pulmonary pressure to protect lungs and high systemic pressure for rapid delivery. Fish have a single circulation. [2].

12. (a) Cardiac muscle may die and heart output fall [2]. (b) Reduced oxygen delivery limits aerobic respiration in cardiac muscle [3]. (c) Accept the biologically justified opposite trend when the causal factor is reversed [2]. (d) One valid assumption, such as other relevant conditions remaining constant [1].

13. (a) $168 - 68 = 100$ [2]. (b) The values change from 68 to 168, with the highest value 168. [2]. (c) Exercise increases oxygen and glucose demand and carbon dioxide removal [3]. (d) Only three conditions / no repeats / small dataset / uncontrolled variation [1].

14. (a) Time after exercise [1]. (b) Pulse rate [1]. (c) Same exercise, duration, participant posture and measurement interval [2]. (d) Use several values across a sensible range, repeat each condition, identify anomalies and calculate means [3]. (e) Repeat on several days and calculate a mean [1].

15. (a) a red-cell protein that combines reversibly with oxygen [2]. (b) Award linked points from: Arteries carry blood away at high pressure and have thick elastic muscular walls. Veins return blood at lower pressure, have wider lumens and valves. Capillaries have one-cell-thick walls, narrow lumens and extensive networks for short diffusion distance and large area. [3]. (c) Correct statement: coronary artery means a vessel supplying the heart muscle with oxygenated blood [2]. (d) Any testable relevant biological question [1].

Self-assessment

Classify the first lost mark as definition, biological knowledge, causal link, data use, experimental design or communication.

Evidence used: Cambridge IGCSE Co-ordinated Sciences 0654 syllabus 2025-2027 and June 2026 papers, mark schemes and Principal Examiner Report supplied by the user. All practice questions are newly written.